CMPSC 448 - Project Report

LLM Authorship Classification

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1. Problem

For this project, I tested whether a classifier could tell which model family wrote a response. I used Gemma, OLMo, and Qwen outputs. I compared the prompt, the response, and both together.

2. Method

I used the public Ai2 Dolci DPO Model Response Pool. It has model responses to shared prompts. I selected Gemma 3 4B, OLMo 2 7B, and Qwen 3 no-reasoning 4B so each prompt had an answer from all three.

Data fields: Each row has `LLM_name` for the model family, `LLM_Input` for the prompt, and `LLM_output` for the response. The CSV also records the prompt ID, model configuration, and source.

The final file has 1,800 rows: 600 responses from each family for 600 matched prompts. I kept rows labeled `IF_sft_data_verified_permissive`, left out WildChat and DaringAnteater rows, and removed simple email, phone-number, and redacted-name matches. These checks are basic filters, not a manual privacy review.

The dataset card lists ODC-BY and says model-specific output terms may apply. Check those terms before sharing the collected responses. Dataset card.

Field	What it contains
LLM_name	Gemma, OLMo, or Qwen
LLM_Input	The prompt sent to the model
LLM_output	The model's response

The data CSV is not included in the repository; the preparation script rebuilds it.

1. Keep prompts that appear for all three model configurations.
2. Apply the source and text checks, then select 600 shared prompts.
3. Keep all responses to one prompt in the same data split.

3. Models and training

I split prompt IDs into 64% training, 16% validation, and 20% testing. The test set has 120 prompts and 360 responses. This kept answers to the same prompt out of different splits.

I lowercased the text and split it into words and punctuation. The vocabulary came from training text only and held up to 12,000 tokens. Both models used 64-dimension embeddings, dropout of 0.3, Adam with a 0.001 learning rate, and four training epochs. Inputs were limited to 128 tokens. For prompt plus response, I reserved 64 tokens for each. I used seed 42.

Model	Main layer	Output
CNN	96 filters, kernel size 3	Max pooling
RNN	Bidirectional LSTM, 64 hidden units	Join both directions

The best checkpoint was the one with the lowest validation loss. I report accuracy and macro-F1. With three balanced families, chance accuracy is 33.3%.

4. RQ1: Can a response identify its model family?

Question: Can a CNN or LSTM identify the model from the response alone?

Answer: Yes, both models scored above the 33.3% chance level. The response-only CNN reached 58.6% accuracy and 58.6% macro-F1. The LSTM reached 51.9% accuracy and 51.8% macro-F1.

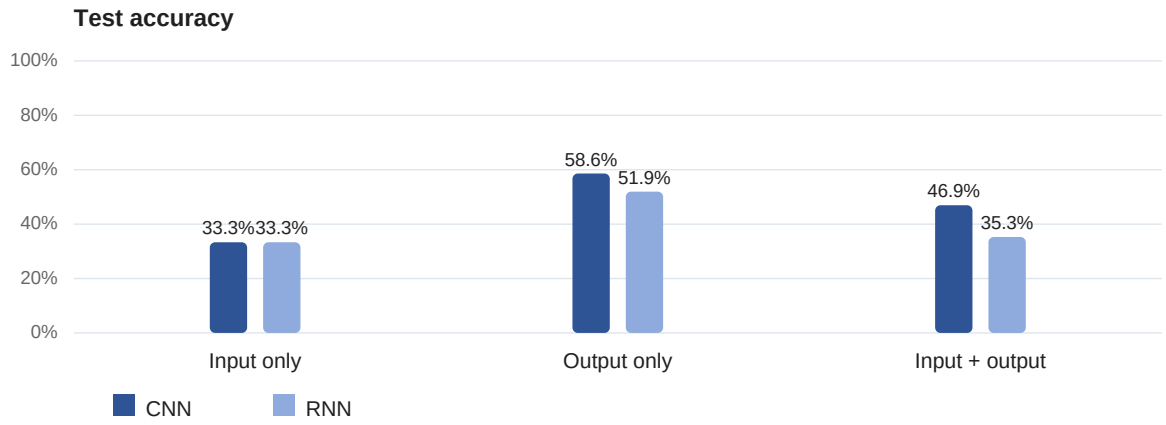

Figure 1. Test accuracy for both models and all three input settings.

Input	CNN accuracy	CNN macro-F1	LSTM accuracy	LSTM macro-F1
Prompt	33.3%	32.7%	33.3%	33.1%
Response	58.6%	58.6%	51.9%	51.8%
Prompt + response	46.9%	42.8%	35.3%	35.3%

5. RQ2: Does the prompt help identify the model?

Question: Can the prompt identify the model before a response is seen? Does it improve the result when added to the response?

Answer: No, not in this experiment. Prompt-only accuracy was 33.3% for both models, equal to chance. Adding the prompt also lowered accuracy: the CNN went from 58.6% with responses alone to 46.9%, and the LSTM went from 51.9% to 35.3%. All three model families answered the same selected prompts, so the prompt alone did not identify the family.

The combined model saw only 64 response tokens because its 128-token limit was split evenly with the prompt. That may help explain why the combined setting did worse than response-only.

Output-only confusion matrices

CNN	Gemma	OLMo	Qwen
Gemma	77	23	20
OLMo	18	71	31
Qwen	22	35	63

RNN	Gemma	OLMo	Qwen
Gemma	63	30	27
OLMo	9	76	35
Qwen	24	48	48

Figure 2. Rows show the actual family; columns show the prediction.

6. RQ4: Which response features differ by model family?

Question: Do response length, vocabulary, punctuation, or formatting differ?

Answer: Average response length and the measured style features differ. Qwen responses were longest at 381 words on average; OLMo responses were shortest at 224 words and had the highest word-variety ratio. Gemma had the highest punctuation rate. These measurements do not show which features caused the classifiers' predictions.

Family	Words	Word variety	Punctuation	Lines
Gemma	354.2	0.594	0.090	29.3
OLMo	223.7	0.627	0.085	19.1
Qwen	381.2	0.587	0.082	35.1

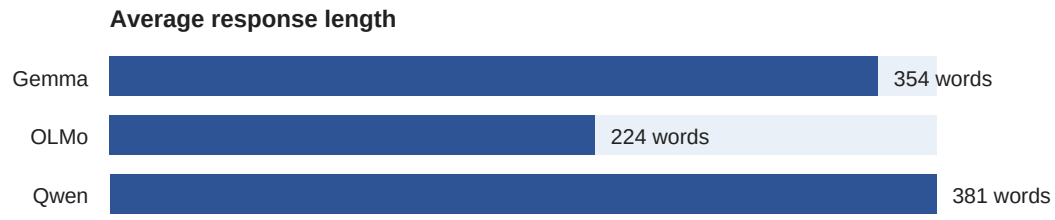

Figure 3. Mean response length in words.

Word variety is the number of unique lowercase words divided by the total words. Punctuation is punctuation characters divided by all response characters. Lines are counted from line breaks.

Qwen averaged 381 words; OLMo averaged 224 and had the highest word-variety ratio. Gemma had the highest punctuation rate. These averages do not describe every response.

7. Challenges and how I addressed them

The source included rows from several collections, so I used the verified permissive category and simple text checks for the dataset. Those checks cannot catch every sensitive detail. I also matched prompts across model configurations and kept each prompt group in one split to reduce leakage.

Long responses had to be cut to fit the 128-token input. In the combined setting, half the space went to the prompt. A larger input limit or a separate experiment with more response tokens could change those results.

8. Key lessons learned

- The response text was more useful than the prompt for identifying the model.
- Grouping by prompt gave a fairer test because the same question did not appear in training and testing.
- The CNN did better than the LSTM on response-only text in this run.
- One split and one configuration per family are not enough to generalize to every model version.

9. Conclusion

The response-only CNN identified the model family better than chance, but its accuracy was 58.6%. The prompt alone did not help, and adding it lowered performance in this setup. The results apply to these three configurations and this test split.

References

Ai2, *Dolci DPO Model Response Pool*, dataset card. Open the dataset card. Accessed October 11, 2026. The card lists ODC-BY and notes that model-specific output terms may apply.

Team Olmo et al. (2025), *Olmo 3*, technical report, arXiv:2512.13961.